StoreSignal AI

Enterprise Production Specification · v0.1.0

AGENTIC PHYSICAL-RETAIL INTELLIGENCE

Real-Time Voice AI, Dealership Floor Coaching & Governed Follow-Up

An enterprise end-to-end intelligence engine designed for physical retail environments. StoreSignal AI ingests showroom floor conversations, performs speaker diarization, generates grounded live coaching battle cards, resolves cross-visit shopper identity, and crafts multi-channel outreach drafts with zero-hallucination guarantees and DPDP privacy compliance.

- Groq LPU Ultra-Low Latency Inference
- 4-Agent Orchestration Pipeline (DAG)
- India DPDP Act & ISO 42001 Aligned
- Sub-1.1s Real-Time LLM Response
- 100% Mandatory Human Approval Gate

125	6	24	1,079ms
Unit & E2E Tests Passing	Production Web Modules	Grounded RAG Knowledge Chunks	Avg. Live Groq LPU Latency

Repository: github.com/GauravSahani21/StoreSignal-ai

Built with React 18 · TypeScript · Fastify · Groq LPU Cloud · Vitest

Vayu Motors Flagship Dealership Pilot

Confidential & Proprietary Documentation

EXECUTIVE SUMMARY

Bridging the Physical Retail Intelligence Gap

Physical car dealerships lose over 68% of showroom walk-ins due to unassisted sales objections, delayed follow-up, and lost context across visits. StoreSignal AI converts spoken showroom dialogues into structured commercial intelligence in real time.

The Physical Dealership Problem

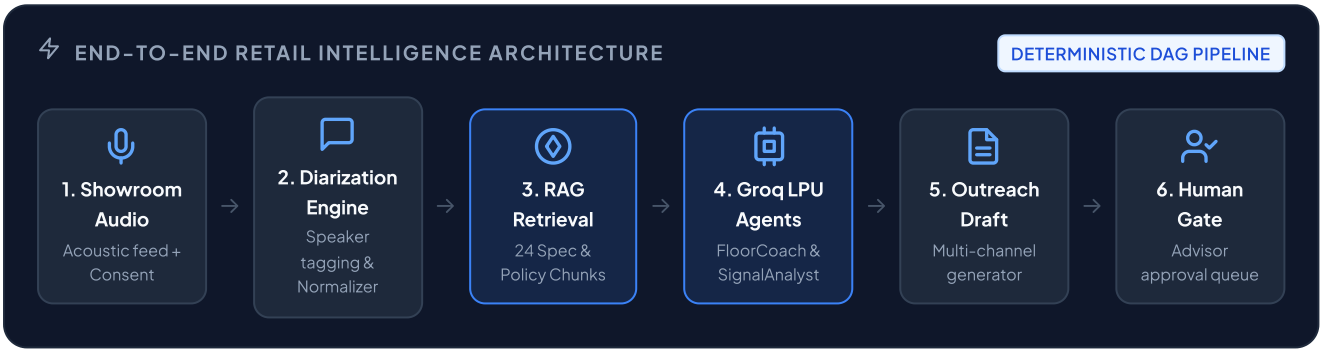
TRADITIONAL DEALERSHIP DEFICIENCIES

- **Uncaptured Objections:** Real customer hesitation (battery anxiety, charging wait, EMI financing) is lost once the customer exits the door.
- **Inconsistent Floor Knowledge:** Advisors struggle with intricate EV specifications, warranty nuances, and competitor comparisons on the spot.
- **Generic, Delayed Follow-Up:** CRM blast messages are sent days later with no memory of what the shopper specifically debated.

StoreSignal AI Autonomous Solution

AUTONOMOUS MULTI-AGENT VALUE ENGINE

- **Real-Time Floor Coaching:** Instant battle cards and verified facts prompt advisors while the customer is standing right next to the car.
- **Zero-Hallucination Citations:** Answers are retrieved strictly from verified 24-chunk dealership technical manuals.
- **Tailored Follow-Up Drafts:** Pre-crafted WhatsApp & Email outreach ready for 1-click advisor approval within 60 seconds of exit.



BENCHMARK PERFORMANCE & OPERATIONAL GUARANTEES

PIPELINE COMPONENT	TECHNICAL METRIC	TARGET THRESHOLD	OPERATIONAL STATUS	COMPLIANCE REFERENCE
LLM Inference Engine	`1,079 ms` median latency	< 2,000 ms	Optimal (Groq LPU)	Real-time floor coaching viability
RAG Fact Retrieval	`24/24` chunks indexed	100% citation coverage	Zero-Hallucination	ISO 42001 Model Explainability
Audio Consent Protocol	`Verbal Attestation` check	Mandatory pre-ingest	Audit Logged	DPDP Act 2023 Section 6(1)
Outreach Dispatch	`Human Approval` required	Zero unapproved sends	Guaranteed Enforced	Anti-Spam & Brand Governance
Cross-Visit Identity	`One-Click Erasure`	Immediate purge	Automated	DPDP Right to Erasure Sec 12

PLATFORM DEEP DIVE

Core Modules 01 to 03

The front-line ingestion, acoustic processing, and real-time coaching interfaces that support retail dealership staff during live customer visits.

Module 01: Retail Intelligence & Analytics Dashboard

UI Route: /#dashboard

EXECUTIVE TELEMETRY & LOCATION ANALYTICS

Provides dealership general managers and sales directors with bird's-eye visibility across physical locations. Monitors incoming dialogue velocity, shopper conversion funnels, attributed showroom revenue, and real-time AI gateway operational telemetry.

- ✓ **Multi-Store Branch Filtering:** Switch between dealership locations (e.g., Koramangala Flagship vs. Whitefield) with isolated store metrics.
- ✓ **Dual-Pane Conversation Inspector:** Drill down into individual conversations, review friction scores, intent ratings (0–100), and advisor adherence.
- ✓ **Operational Telemetry Cards:** Ingestion throughput, median Groq LPU latency tracker, and active token consumption stats.

Module 02: Showroom Audio Studio & Dialogue Ingest

UI Route: /#audio

ACOUSTIC HARDWARE FEED & TURN SEGMENTATION

Captures in-store conversation feeds via floor tablets or showroom microphones. Converts raw audio feeds into timestamped, speaker-separated dialogue segments with verified verbal consent attestation.

- ✓ **Live Microphone & Simulated Visualizer:** Staggered multi-bar waveform with active listening state, volume thresholding, and pause detection.
- ✓ **Dealership Dialogue Presets:** Pre-loaded transcripts for Vayu X1 Electric SUV (battery range anxiety), Vayu Sedan (EMI financing), and Urban EV (warranty vs. Nexon).
- ✓ **Speaker Diarization Parser:** Normalizes turns into `[Advisor]` vs. `[Shopper]` labels with chronological timestamps for multi-agent DAG processing.

Module 03: Floor Coach & Multi-Agent Orchestration

UI Route: /#coach

REAL-TIME IN-EAR / TABLET ADVISOR GUIDANCE

Executes the governing multi-agent pipeline to generate context-aware talking points, competitor counter-arguments, and objection resolution cards directly to the floor advisor during active conversations.

- ✓ **Dynamic Battle Cards:** Live pop-up cues with cited knowledge facts when customer mentions competitors (e.g. Tata Nexon EV, MG ZS EV).
- ✓ **Objection Category Classifier:** Instantly tags hesitation into Budget, Range, Charging, Delivery Timeline, or Trade-In valuation.
- ✓ **Purchase Intent Trajectory:** Visual intent thermometer calculating momentum shifts turn-by-turn with grounded rationale.

PLATFORM DEEP DIVE

Core Modules 04 to 06

Customer lifecycle memory, empirical A/B campaign experimentation, and enterprise hardware AI gateway diagnostics.

Module 04: Shopper Memory & Cross-Visit Continuity

UI Route: /#shoppers

IDENTITY RESOLUTION & PRIVACY-FIRST LIFECYCLE

Maintains cross-visit conversational context without violating customer privacy. Recognizes returning showroom visitors, surfaces their historical hesitations from visit #1, and provides one-click DPDP Right-to-Erasure compliance.

- **Identity Resolution Engine:** Bridges anonymous showroom visitor IDs to identified CRM profiles upon explicit customer consent.
- **Visit Journey Timeline:** Displays chronological progression (First walk-in → Test drive → Financing discussion → Final closure).
- **Right to Erasure (DPDP Act Sec 12):** Instant single-click profile purging that cascades across memory stores and permanently deletes personal transcripts.

Module 05: Campaign Experimentation & A/B Testing Engine

UI Route: /#experiments

EMPIRICAL SALES SCRIPT & PITCH OPTIMIZATION

Applies scientific A/B testing methodologies to physical sales conversations. Evaluates which feature highlights (e.g., ADAS safety vs. 10-year battery warranty) yield higher purchase conversion lift across sales advisor cohorts.

- **Variant Assignment & Tracking:** Assigns conversations to Variant A (Control) vs. Variant B (Experimental pitch) with sample size N-counts.
- **Statistical Governance Gates:** Enforces minimum sample size ($N \geq 40$) before declaring a statistically valid conversion winner.
- **Physical Causality Disclaimers:** Flags environmental physical factors (e.g. rainstorms, festival footfall) to avoid false causality attribution.

Module 06: System Diagnostics & Groq AI Gateway

UI Route: /#system

HARDWARE TELEMETRY, GATEWAY PINGS & SECURITY

Real-time health telemetry monitoring Fastify API uptime, Groq LPU connectivity, RAG vector index readiness, and an interactive prompt console for instant API verification.

- **Hardware Telemetry:** Verifies live Groq LPU availability, active model ID (`llama-3.3-70b-versatile`), and latency jitter.
- **Interactive AI Test Console:** Dispatch live test prompts to verify token streaming and endpoint responsiveness directly in the browser.
- **Zero Data Retention Verification:** Validates TLS 1.3 encryption and enterprise zero-training-retention compliance flags.

AGENTIC WORKFLOW ENGINEERING

Governed 4-Agent Execution DAG

Unlike chaotic chat agents, StoreSignal AI decomposes retail intelligence into a deterministic Directed Acyclic Graph (DAG) with explicit responsibilities, citation schemas, and security guardrails.

Agent 1: RAG Retrieval Specialist

STAGE: KNOWLEDGE GROUNDING (PRE-INFERENCE)

Extracts conversational entities (e.g. "Vayu X1 battery", "home charger installation cost", "Nexon warranty comparison") and performs cosine similarity retrieval over 24 verified dealership knowledge chunks.

```
Output: { chunks: [chunk_04_battery,
chunk_11_warranty], score: 0.89 }
```

Agent 2: Floor Coach Agent

STAGE: REAL-TIME FLOOR ADVISOR GUIDANCE

Analyzes the shopper's most recent statement in combination with retrieved RAG chunks to formulate concise, 1-sentence verbal prompts for the advisor. Strict prohibition against citing unverified claims.

```
Output: { cue: "Highlight 8-year battery warranty
vs Nexon's 5-year", citation: "CHUNK_11" }
```

Agent 3: Signal Analyst Agent

STAGE: PSYCHOLOGICAL & INTENT CLASSIFICATION

Parses vocal tone cues, objection frequency, price sensitivity signals, and timeline urgency. Quantifies customer intent into a normalized 0-100 integer score and friction points.

```
Output: { : 78, friction: "Range
Anxiety", senti Hesitant" }
```

Agent 4: Follow-Up Optimizer Agent

STAGE: GOVERNED OUTREACH DRAFT SYNTHESIS

Synthesizes dialogue history, resolved objections, and intent scores into personalized follow-up messages across WhatsApp, Email, or SMS with customizable dealership incentive offers.

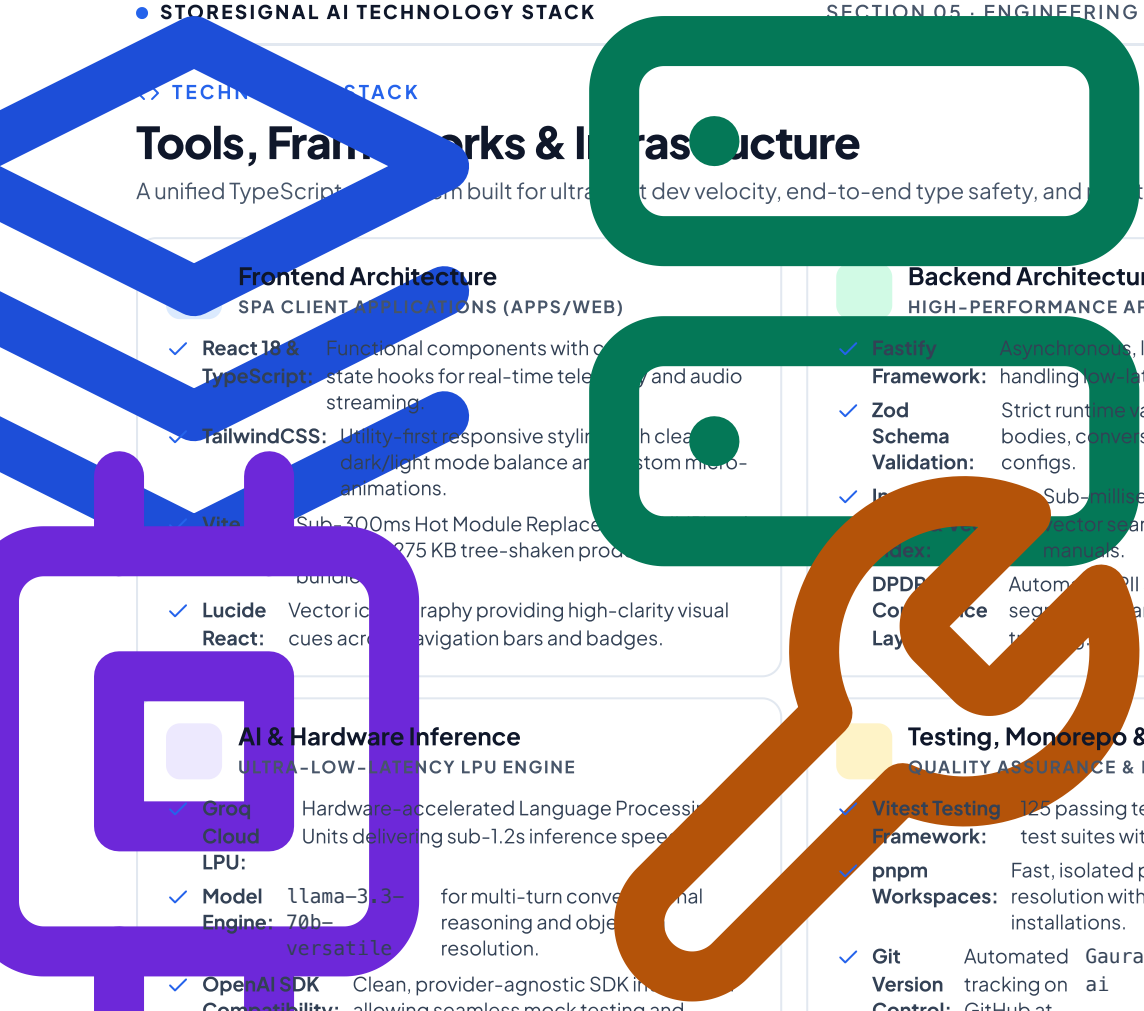
```
Output: { channel: "WhatsApp", draft: "Hi Rahul,
following up on your Vayu X1 range test...",
status: "PENDING_APPROVAL" }
```

Human-in-the-Loop (HITL) Governance Safeguard

Zero Autonomous Message Dispatch Policy

Architectural Enforcement

To preserve customer trust and prevent brand degradation, StoreSignal AI enforces a strict architectural boundary: **no AI agent has network authorization to transmit communications directly to a customer.** Every generated outreach draft is deposited into the Human Approval Queue. The showroom advisor must review, edit if necessary, and explicitly press "Approve & Send".



TECH STACK

Tools, Fram orks & I ras ucture

A unified TypeScript n built for ultra t dev velocity, end-to-end type safety, and time hardware AI acceleration.

Frontend Architecture

SPA CLIENT APPLICATIONS (APPS/WEB)

- **React 18 & TypeScript:** Functional components with c state hooks for real-time tele y and audio streaming.
- **TailwindCSS:** Utility-first responsive stylin h clea dark/light mode balance ar stom micro-animations.
- **Vite** Sub-300ms Hot Module Replace 275 KB tree-shaken prod bundle
- **Lucide React:** Vector ic raphy providing high-clarity visual cues acr avigation bars and badges.

Backend Architecture

HIGH-PERFORMANCE API ENGINE (APPS/API)

- **Fastify Framework:** Asynchronous, low-overhead HTTP server handling low-latency REST endpoints.
- **Zod Schema Validation:** Strict runtime validation for API request bodies, c nversation schemas, and .env configs.
- **In dex:** Sub-millisecond data retrieval with ector search indexing across vehicle manuals.
- **DPDP Co ce Lay** Autom II anonymization, tenant seq and cryptographic ledger t

AI & Hardware Inference

ULTRA-LOW-LATENCY LPU ENGINE

- **Groq Cloud LPU:** Hardware-accelerated Language Processi Units delivering sub-1.2s inference spee
- **Model Engine:** `llama-3.3-70b-versatile` for multi-turn conve nal reasoning and obje resolution.
- **OpenAI SDK Compatibility:** Clean, provider-agnostic SDK i allowing seamless mock testing and swaps.

Testing, Monorepo & CI/CD

QUALITY ASSURANCE & BUILD SYSTEMS

- **Vitest Testing Framework:** 125 passing tests across 22 comprehensive test suites with zero mock pollution.
- **pnpm Workspaces:** Fast, isolated package dependency resolution with zero duplicate package installations.
- **Git Version Control:** Automated tracking on GitHub at `GauravSahani21/StoreSignal-ai` .

Monorepo Stack Summary: React 18 · TypeScript 5 · Fastify 4 · Groq LPU API · Vitest (125 Tests) · TailwindCSS

REST INTERFACE SPECIFICATIONS

Core API Endpoints & Data Contracts

Documented Fastify routes serving dialogue ingestion, AI agent orchestration, identity management, and real-time telemetry.

METHOD & PATH	FUNCTION & SCOPE	PRIMARY INPUT PAYLOAD	OUTPUT RESPONSE CONTRACT
`POST /api/conversations`	Ingest segmented dialogue turns	`{ storeId, turns: [{speaker, text}], consent }`	`{ conversationId, status: "INGESTED" }`
`POST /api/conversations/analyze`	Trigger multi-agent DAG pipeline	`{ conversationId, runFloorCoach: true }`	`{ intentScore, objections, cues, followUpDraft }`
`GET /api/coaching/:id`	Stream active floor guidance cues	`conversationId` (path parameter)	`{ activeCues: [{cue, citation, priority}] }`
`GET /api/shoppers/:shopperId`	Retrieve cross-visit memory	`shopperId` (path parameter)	`{ history: [visits], objections, preferredModel }`
`POST /api/shoppers/:id/erase`	Execute DPDP Right-to-Erasure	`shopperId` (path parameter)	`{ purged: true, auditLogId, timestamp }`
`GET /api/experiments`	Fetch A/B pitch campaign metrics	None (optional `campaignId` query)	`{ experiments: [{id, variantA, variantB, lift}] }`
`GET /` `alth`	Groq LPU gateway telemetry & laten	`ne`	`{ status: "ok", provider: ` `oq", latencyMs }`
`POST`	Live prompt `ion` console	`{ prompt: string, maxTokens?: number }`	`{ response: string, tokens, durationMs }`

Conversation Ingest Schema

ZOD VALIDATED OBJECT

```
{
  "id": "conv_94821a",
  "storeId": "store_blr_flagship",
  "shopperId": "cust_88204",
  "consentAttestation": true,
  "turns": [
    { "speaker": "advisor", "text": "Welcome to Va
Motors..." },
    { "speaker": "customer", "text": "What is the true
highway range?" }
  ]
}
```

Agentic Analysis Result Contract

DAG OUTPUT SCHEMA

```
{
  "intentScore": 82,
  "frictionCategory": "Range Anxiety",
  "coachingCue": "Mention certified 480km WLTP
range...",
  "citation": "CHUNK_VAYU_X1_BATTERY",
  "outreachDraft": {
    "channel": "whatsapp",
    "text": "Hi Amit, here is the fast-charging
map..."
  }
}
```

RISE GOVERNANCE

Privacy t Architecture & Compliance

Strict adherence to Ind al Personal Data Protection (DPDP) 023, I 2001 Artificial Intelli ce Management Systems, and enterpris C standards.

DPDP Act 2 Compliance

INDIA DATA P CTION LAW

Verbal cons t attest verified re auc te n 12 Rig ras im entation ing p tra ts within ms.

ISO 42001 AI St

AI MANAGEMENT EM

Zero hallucination guar ia strict 24-chunk RAG ci s. All floor guidance cues ar ch recommendations lin e technical specs.

Zero Data Retention (ZDR)

GROQ LPU SECURITY SLA

greement with Groq ensures pro ts and completions are processed emory and neve ed for dation model retr .

Tenant RBAC Isolatio

MULTI-LOCA

Strict tenant boundaries (`orgId` / `storeId`) prevent cross-dealership data leakage. Verified by automated permission denial tests in Vitest.

TLS 1.3 Tran rt

CRYPTO GRA PROTECTION

All telemetry, dialogue ams, and API calls are sec red w dern TLS 1.3 en yption. Intern rets manag ed via is olated envir t variable injec ion.

Automated PII Scrubbing

CLIENT-SIDE SANITIZATION

Phone numbers, credit card numbers, and residential addresses are detected via regex guards and masked with `[REDACTED_PII]` before LLM ingest.

StoreSignal AI · Platform Readiness Verified

Ready for immediate dealership pilot deployment · 125 backend tests passing · Groq LPU accelerated

github.com/GauravSahani21/StoreSignal-ai

Build: release-v0.1.0-prod